

BOOTSTRAP BASICS

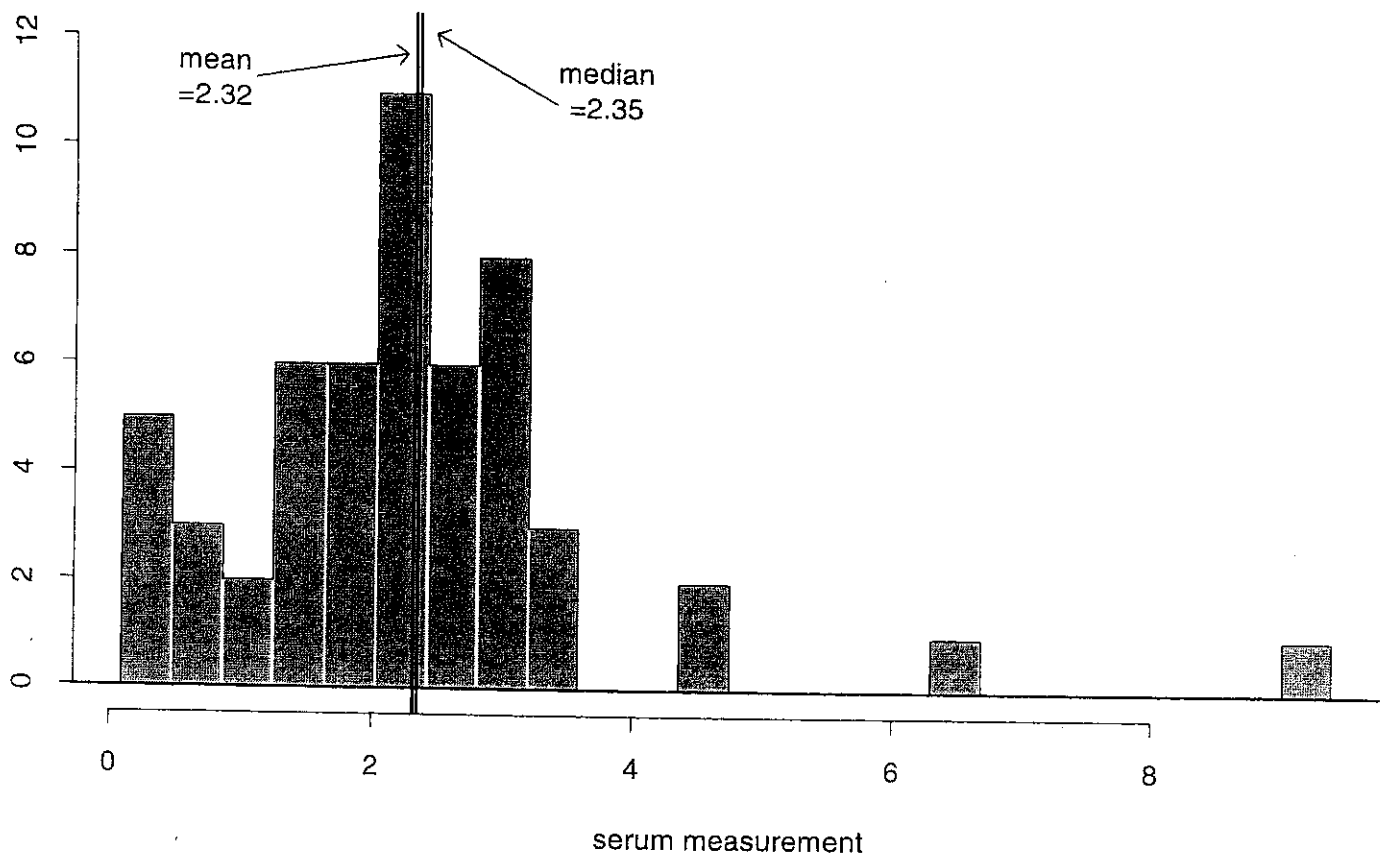
Bradley Efron

Stanford

B. Efron and R. Tibshirani:

“An Introduction to the Bootstrap”

Serum measurements for 54 metastatic patients



The serum data:

```
[1] 0.1 0.1 0.2 0.4 0.4 0.6 0.8 0.8 0.9 0.9 1.3 1.3 1.4 1.5 1.6 1.6 1.7
[18] 1.7 1.7 1.8 2.0 2.0 2.2 2.2 2.2 2.3 2.3 2.4 2.4 2.4 2.4 2.4 2.4 2.5
[35] 2.5 2.5 2.7 2.7 2.8 2.9 2.9 2.9 3.0 3.1 3.1 3.2 3.2 3.3 3.3 3.5 4.4
[52] 4.5 6.4 9.4
```

A FAMOUS FORMULA

DATA $x_1, x_2, x_3 \dots x_n$ $[n = 54]$

MEAN $\hat{\theta} = \sum_{i=1}^n x_i/n$ estimates the expected
value of a random x $[= 2.32]$

FAMOUS FORMULA

$$\boxed{\text{sterr} = \sqrt{\sum (x_i - \hat{\theta})^2 / n^2}} \quad [= .206]$$

says how well $\hat{\theta}$ estimates the true expectation

Confidence Interval $\hat{\theta} \pm 1.96 \text{ sterr}$

The Catch No famous formulas for median,
correlation...

BOOTSTRAP A computer-based way of estimating
standard errors, biases, confidence intervals for
any statistic.

BOOTSTRAP RESAMPLING

Given the Data $\mathbf{x}$

A Bootstrap Sample is a random sample of size n taken *with* replacement from $\mathbf{x}$,

$$\mathbf{x}^* = (x_1^*, x_2^*, x_3^* \dots, x_n^*)$$

BOOTSTRAP REPLICATION

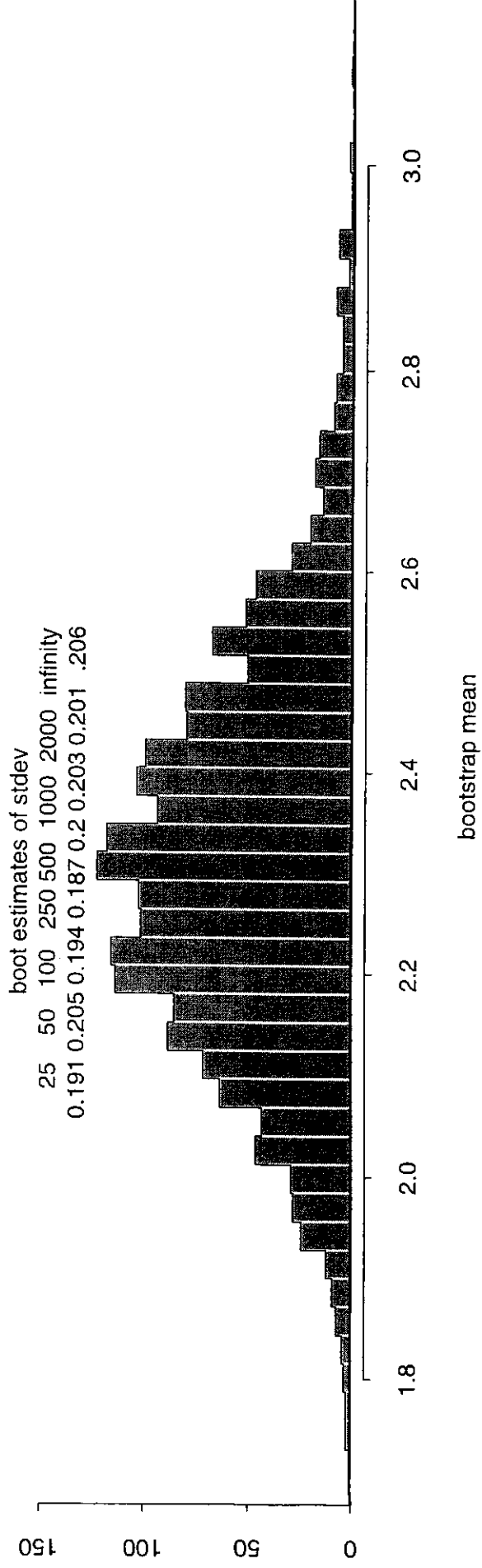
of statistic $\hat{\theta} = s(\mathbf{x})$ is

$$\hat{\theta}^* = s(\mathbf{x}^*)$$

Bootstrap Estimate of Standard Error is the observed standard deviation of repeated bootstrap replications $\hat{\theta}^*$.

Reference Chapter 6 of “An Introduction to the Bootstrap” B. Efron and R. Tibshirani, Chapman & Hall 1993.

2000 bootstrap means for serum data



2000 bootstrap medians

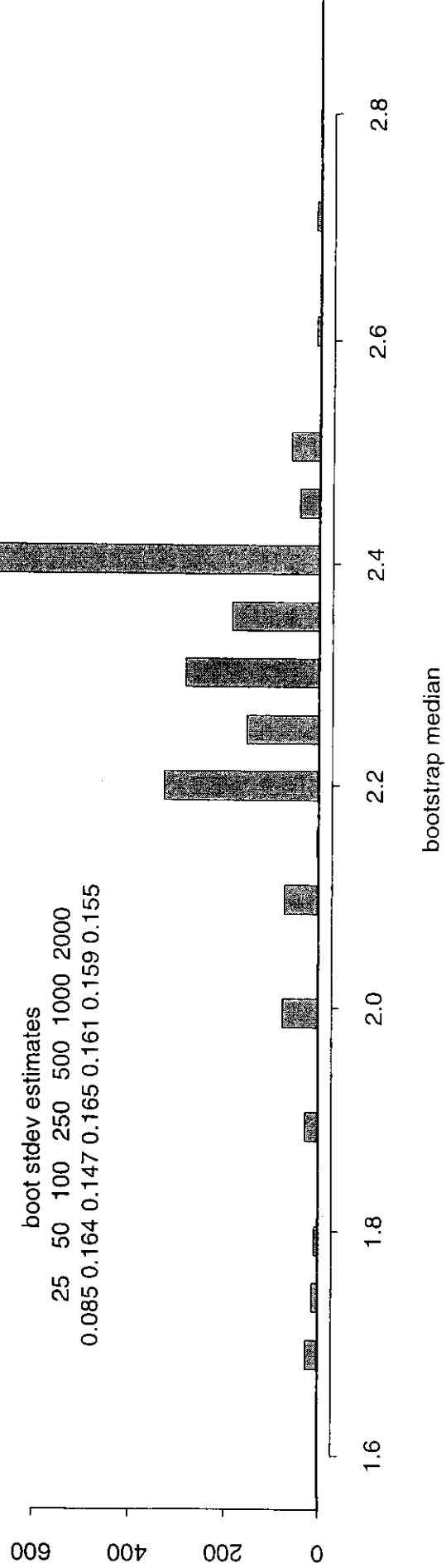
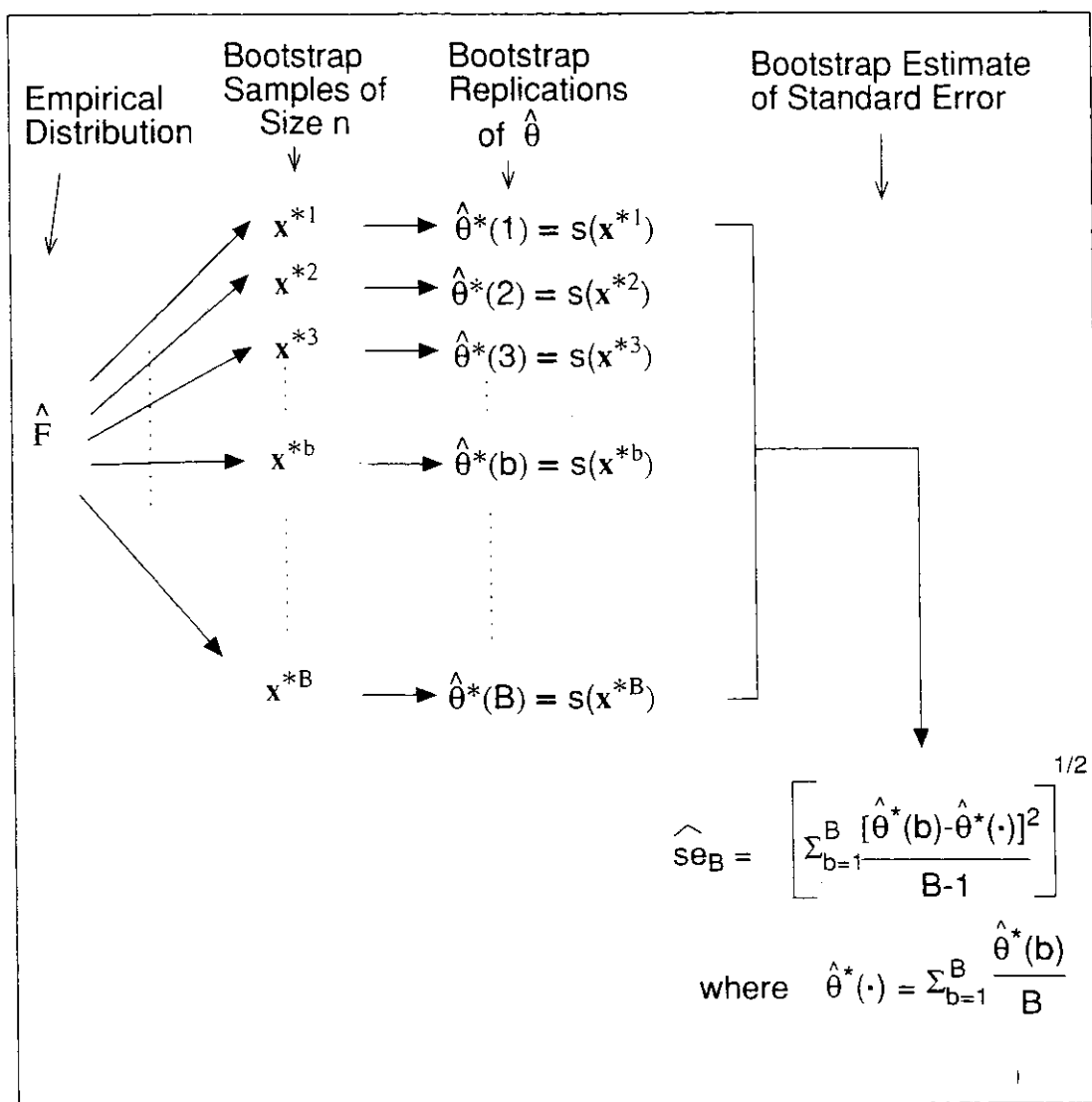


Figure 6.1. The bootstrap algorithm for estimating the standard error of a statistic $\hat{\theta} = s(\mathbf{x})$; each bootstrap sample is an independent random sample of size n from $\hat{F}$. The number of bootstrap replications B for estimating a standard error is usually between 25 and 200. As $B \rightarrow \infty$, $\hat{se}_B$ approaches the plug-in estimate of $se_F(\hat{\theta})$.



A word about notation: in (6.7) we write $se_{\hat{F}}(\hat{\theta}^*)$ rather than $se_{\hat{F}}(\hat{\theta})$ to avoid confusion between $\hat{\theta}$, the value of $s(\mathbf{x})$ based on the observed data, and $\hat{\theta}^* = s(\mathbf{x}^*)$ thought of as a random variable based on the bootstrap sample. The fuller notation $se_{\hat{F}}(\hat{\theta}(\mathbf{x}^*))$ emphasizes that $se_{\hat{F}}$ is a bootstrap standard error: the actual data $\mathbf{x}$ is held fixed in (6.7); the randomness in the calculation comes from the variability of the bootstrap samples $\mathbf{x}^*$, given $\mathbf{x}$. Similarly we will write $E_{\hat{F}}g(\mathbf{x}^*)$ to indicate the *bootstrap expectation* of a func-

BOOTSTRAP PROGRAM IN "SPLUS"

```
function(x, B, stat)
{
#bootstrap stat(x) B times
#x is nxp matrix of data, can be a vector

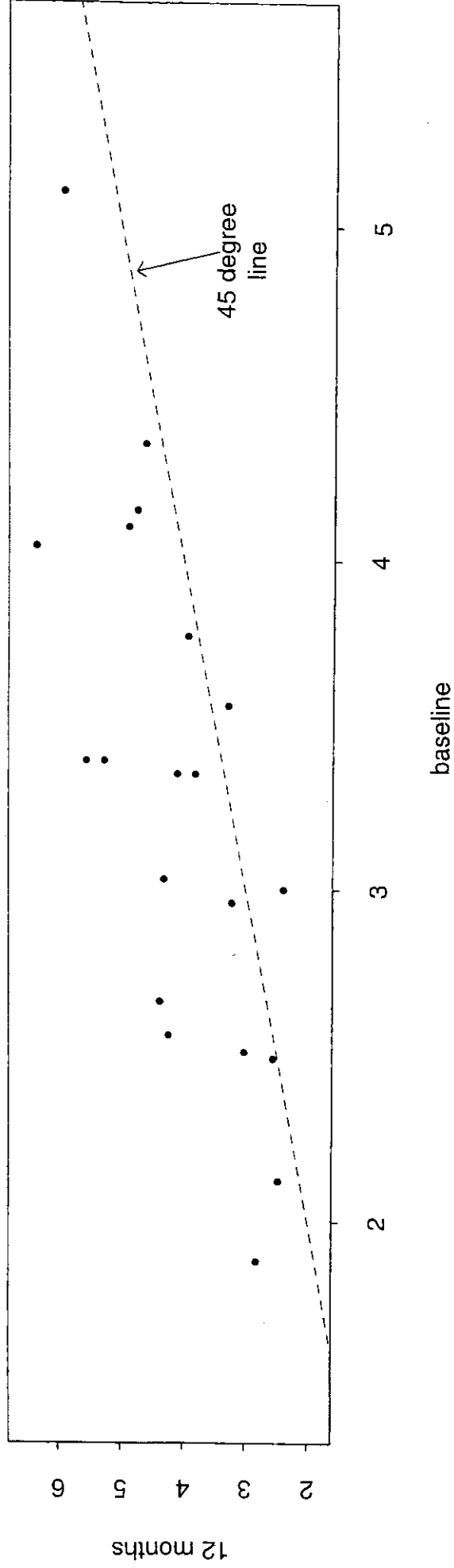
  x <- as.matrix(x)
  n <- nrow(x)
  ss <- rep(0,B)

## choosing the bootstrap samples:
  i <- sample((1:n), n * B, replace = T)
  i <- matrix(i, n)

## doing the bootstrap replications
  for(b in 1:B) {
    xx <- x[i[, b], ]
    ss[b]_stat(xx)
  }

  ss
}
```

Fig 4: cd4 counts for n=20 patients at baseline and after 12 months Sequinavir



2000 nonparametric bootreps of correlation coefficient

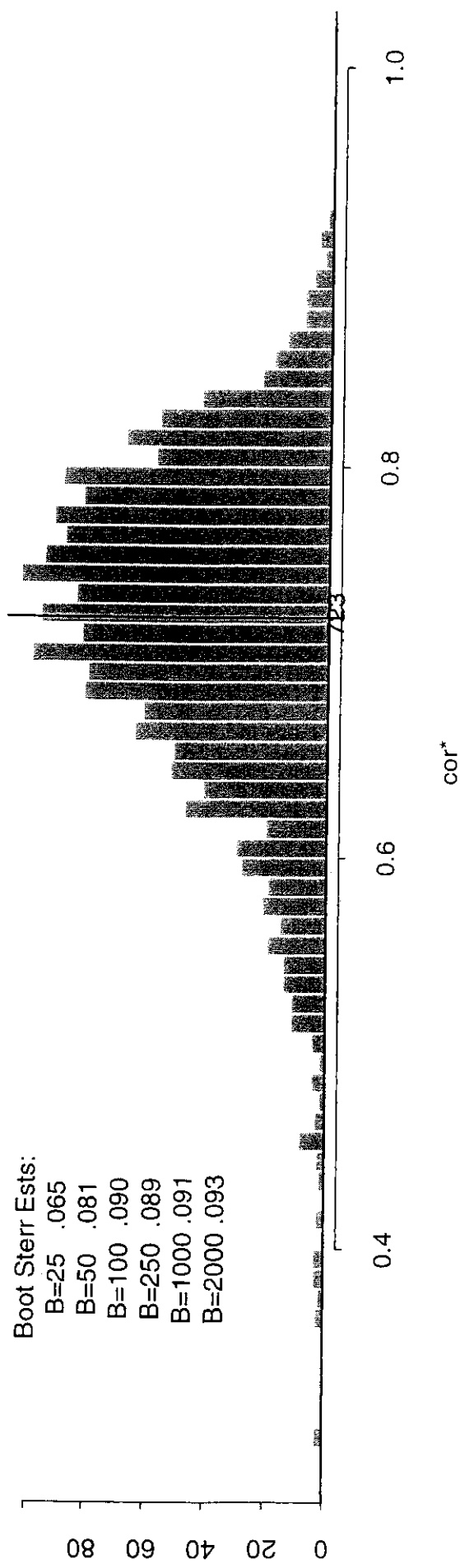


fig1. Kidney data, 157 normal volunteers;
loess(tot~age, span=.5); 8/9/02

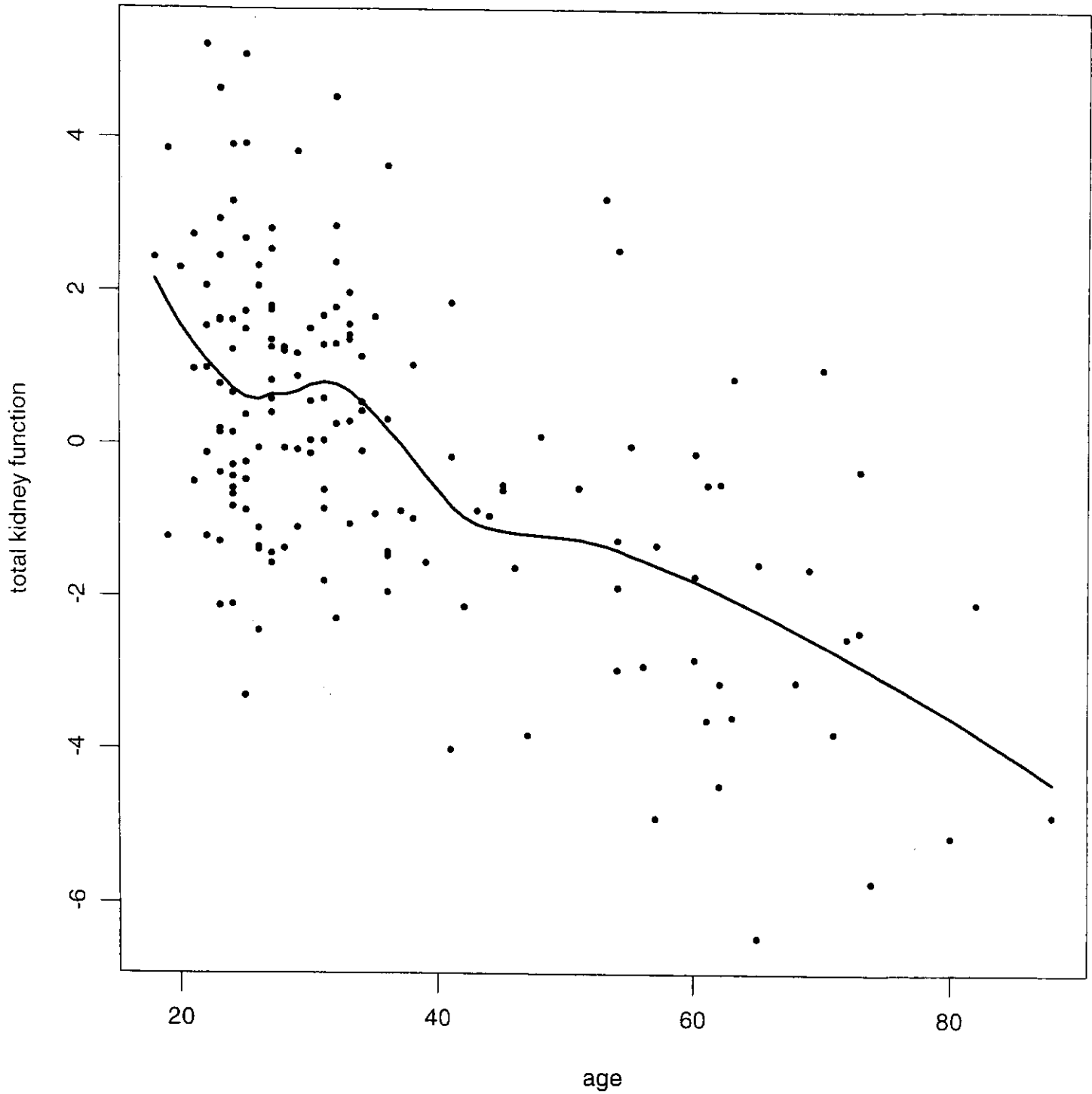
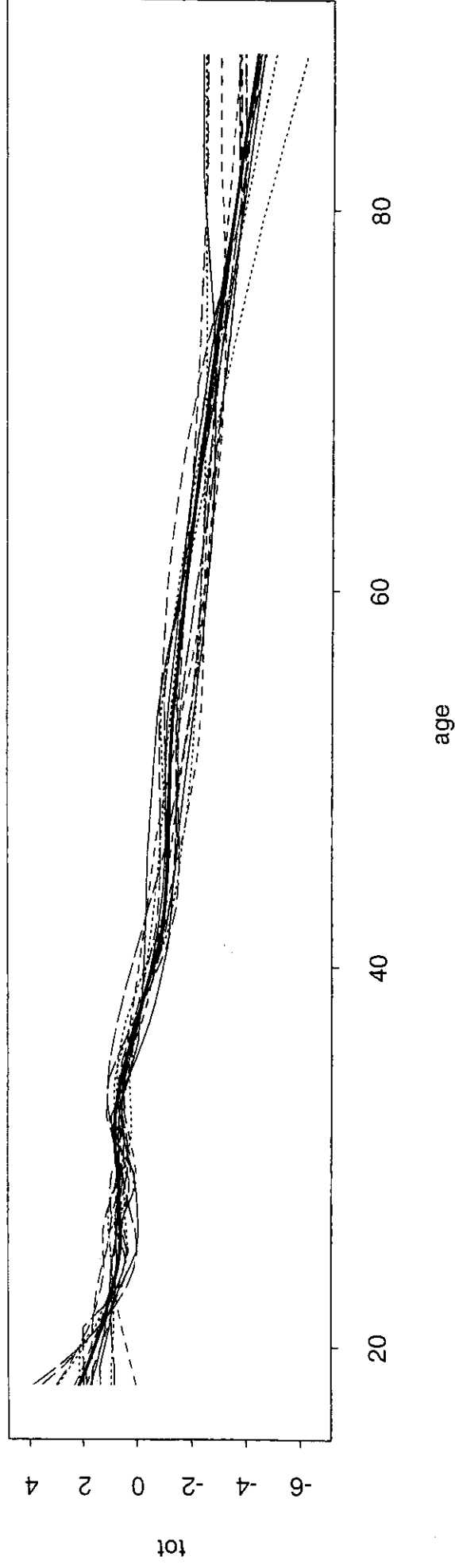


Fig2. first 15 bootstrap replications of
loess Kidney fit



Bootstrap Standard Error estimates,
B=100 bootstrap replications

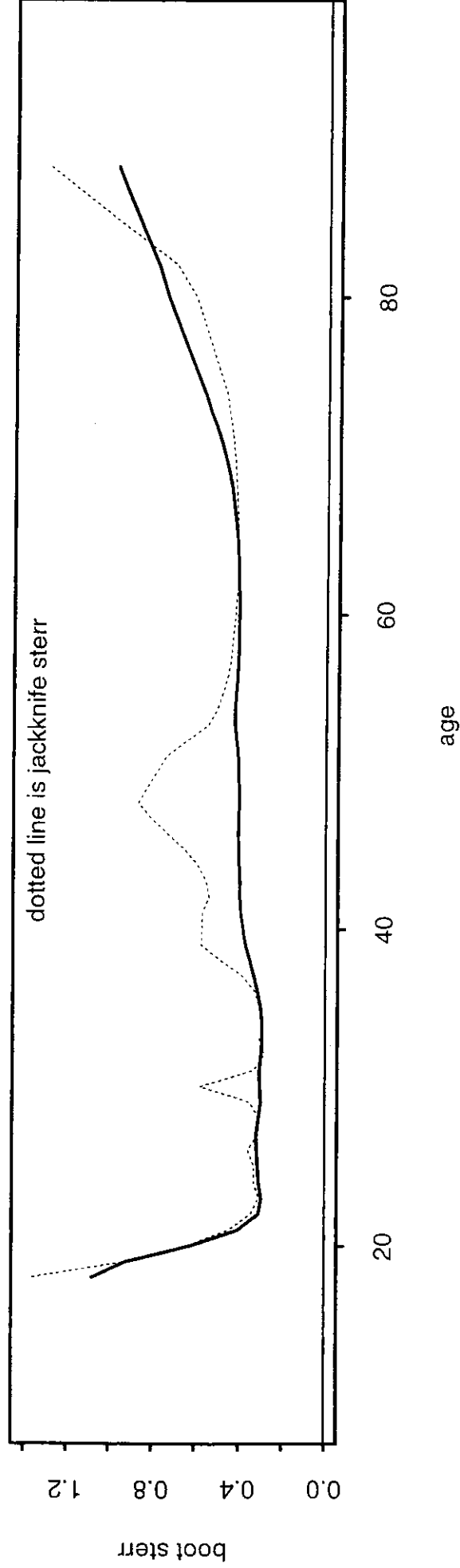
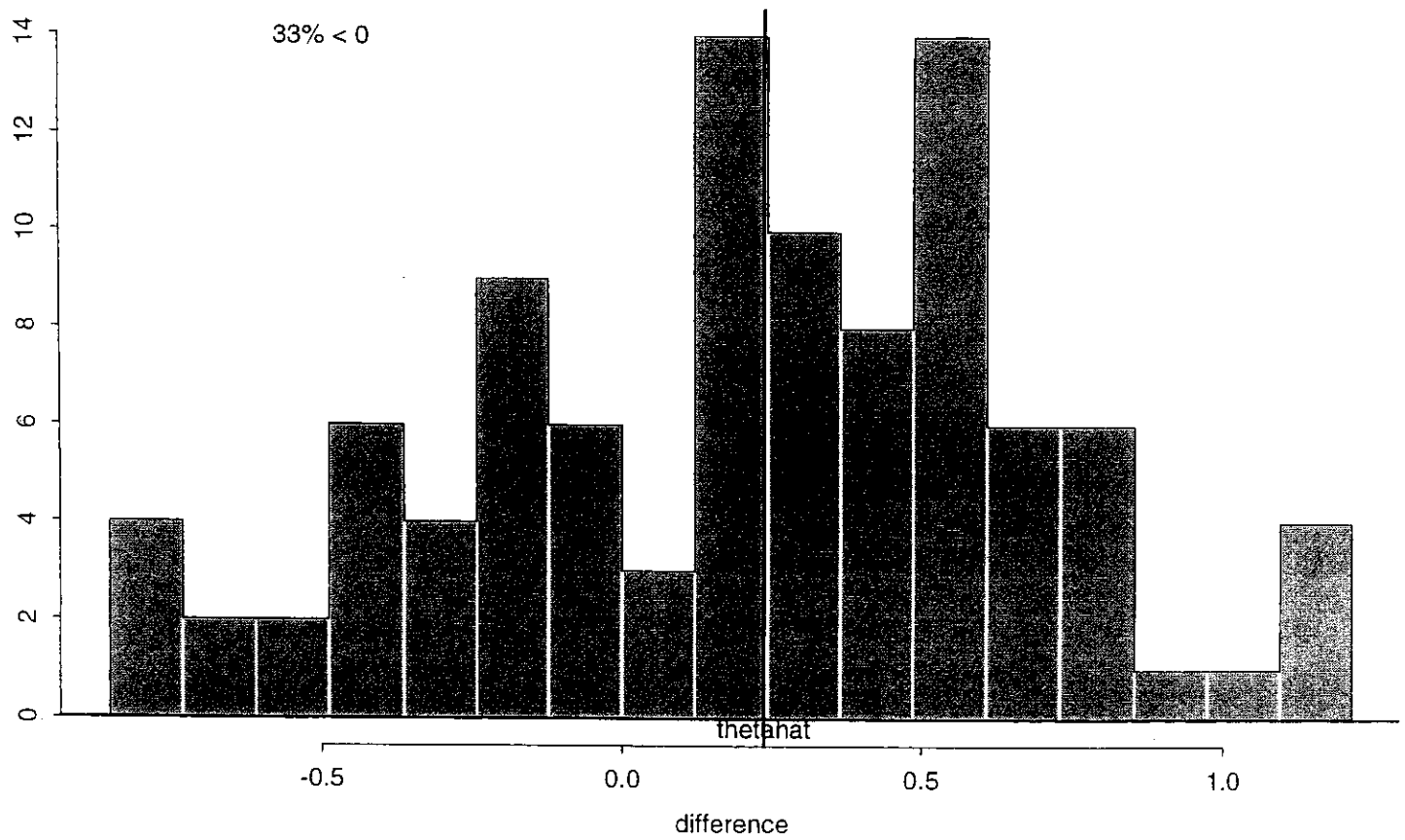


fig3. 100 Bootreps: $\hat{f}(31)^* - \hat{f}(26)^*$



Bootstrap Confidence Intervals

- **Approximate $1 - 2\alpha$ “BCA” Interval**

Requires 2 preliminary calculations:

Bias Constant:

$$“z_0” = \Phi^{-1}(\text{proportion } \hat{\theta}^* \text{'s} \leq \hat{\theta})$$


 standard normal cdf

Acceleration

$$\frac{\frac{\sum_{i=1}^n [\hat{\theta}_{(i)} - \hat{\theta}_{(.)}]^3}{1}}{6 \left\{ \sum_{i=1}^n (\hat{\theta}_{(i)} - \hat{\theta}_{(.)})^2 \right\}^{1/2}} = “a”$$

$$\hat{\theta}_{(i)} = t(x_1, x_2, x_i, x_{i-1}, x_{i+1}, \dots, x_n)$$

BCA Interval is (α_1, α_2) quantiles of boot histogram

$$\alpha_1 = \Phi \left(z_0 + \frac{z_0 + z^{(\alpha)}}{1 - a(z_0 + z^{(\alpha)})} \right)$$

$$\alpha_2 = \Phi \left(z_0 + \frac{z_0 + z^{(1-\alpha)}}{1 - a(z_0 + z^{(1-\alpha)})} \right)$$

$$\alpha = .05 \begin{cases} z^{(\alpha)} = -1.64 \\ z^{(1-\alpha)} = 1.64 \end{cases}$$

Efron & Tibshirani Chapter 14

Favorable Properties

Standard Interval $\hat{\theta} \pm \hat{\sigma} z^{(\alpha)}$ has non-coverage probabilities

$$\alpha \pm O(1/\sqrt{n})$$

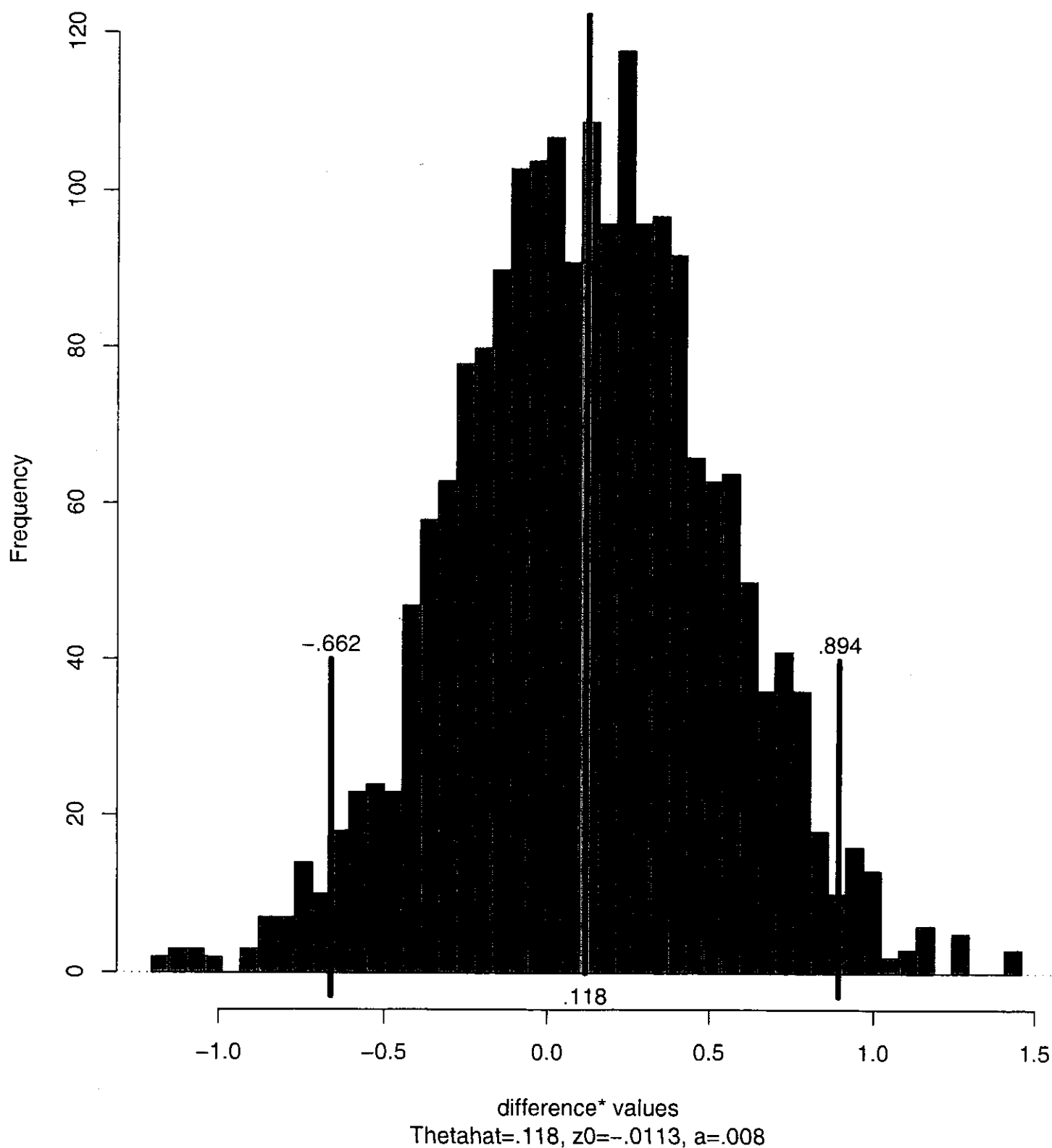
in each tail.

BCA Interval $\alpha \pm O(1/n)$

Transformation Invariance If $\phi = m(\theta)$ monotone transformation,

$$\text{BCA}[\phi] = \phi(\text{BCA}[\theta]).$$

2000 bootstrap reps of age[31]–age[26] fitted values,
kidney data, and the BCA 95% Confidence Limits



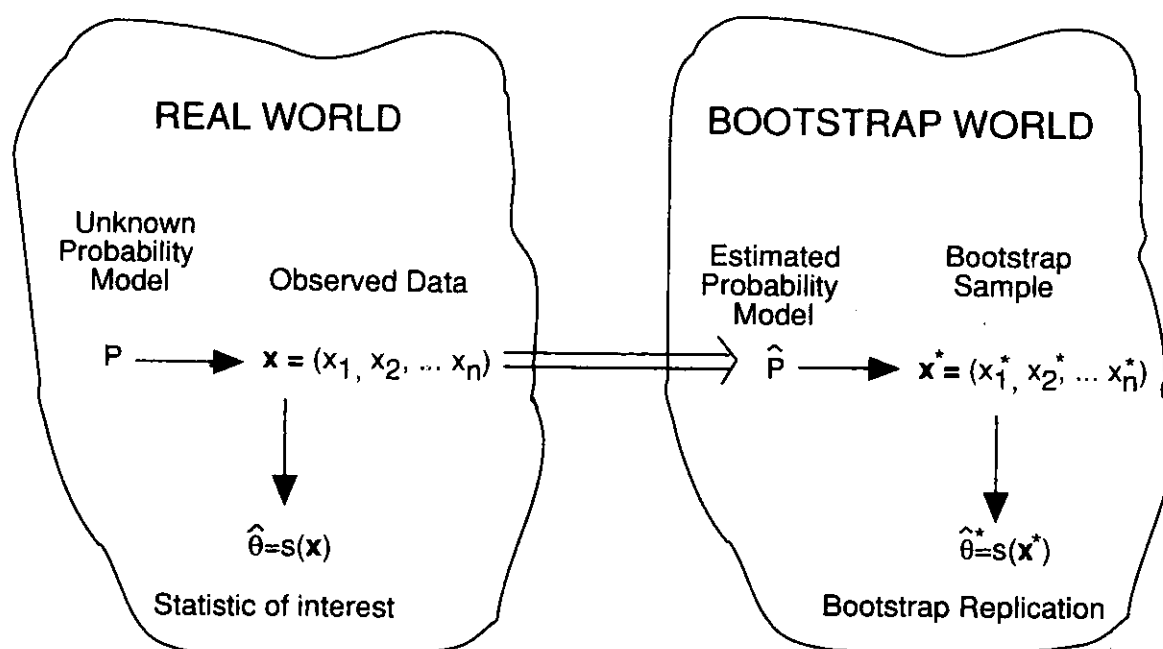


Figure 8.3. Schematic diagram of the bootstrap applied to problems with a general data structure $P \rightarrow \mathbf{x}$. The crucial step " $\Rightarrow$ " produces an estimate $\hat{P}$ of the entire probability mechanism P from the observed data $\mathbf{x}$. The rest of the bootstrap picture is determined by the real world: " $\hat{P} \rightarrow \mathbf{x}^*$ " is the same as " $P \rightarrow \mathbf{x}$ "; the mapping from $\mathbf{x}^* \rightarrow \hat{\theta}^*, s(\mathbf{x}^*)$, is the same as the mapping from $\mathbf{x} \rightarrow \hat{\theta}, s(\mathbf{x})$.

arrow " $\rightarrow$." In specific applications we need to define the arrow more carefully, as in (8.5) for the two-sample problem. The data set $\mathbf{x}$ may no longer be a single vector. It has a form dependent on the data structure, for example $\mathbf{x} = (\mathbf{z}, \mathbf{y})$ in the two-sample problem. Having observed $\mathbf{x}$, we calculate a statistic of interest $\hat{\theta}$ from $\mathbf{x}$ according to the function $s(\cdot)$.

The bootstrap side of Figure 8.3 is defined by the analogous